SDC HACKATHON 2026

PHASE 1 PRELIMINARY PROJECT PROPOSAL

Explainable career pathways for an AI-shaped workforce

TEAM	Fishtail
PROJECT	FutureShift AI
DOMAIN	Domain 1 - Aiversecity Future-Ready Workforce
LEADER	YAO Yunshu
UNIVERSITY	Xiamen University Malaysia
SUBMISSION FILE	SDC2026_Fishtail_Preliminary.pdf

Submission deadline: 25 October 2026

Prepared 24 October 2026

1. Team and Project Overview

Team Name	Fishtail
Selected Domain	Domain 1 - Aiversecity Future-Ready Workforce
Selected SDG Track	SDG 4 - Quality Education (primary target: 4.4)
Project / Solution Name	FutureShift AI
Name (Leader)	YAO Yunshu
Email & Phone	yao18268483040@gmail.com | +60 11-2596 4690
University	Xiamen University Malaysia
Number of Team Members	1

Primary submission links: Live prototype | Public repository

2. Executive Summary & Problem Statement

2.1 The Problem Context

Malaysia's workforce must adapt as AI changes tasks, skill demand and career pathways faster than traditional guidance systems update. The Domain 1 brief reports that 56% of Malaysian professionals use AI tools while only 26% have received formal training, and cites a national estimate that more than 600,000 workers may require reskilling within three to five years. Students, early-career workers and career changers therefore face two linked problems: they often cannot translate their existing skills into realistic adjacent careers, and they struggle to turn a broad recommendation into a credible learning plan. Generic AI advice can amplify this uncertainty when it hides data sources, treats weak proxies as labour demand, or invents qualifications. The result is wasted learning effort, reduced employability and unequal access to future-ready skills. FutureShift AI addresses this gap with an explainable, multilingual workflow that distinguishes personal fit, official occupational evidence, sustainability contribution and learning actions. It is designed for exploration rather than employment prediction, with visible coverage limits and source dates so users can judge what the evidence can and cannot support.

Word count: 173 / 250

2.2 The Proposed AI Solution

FutureShift AI is a deployed Streamlit application that converts a questionnaire or optional CV into a reviewable skills profile. A deterministic scoring engine compares that profile with a quality-controlled O*NET career catalogue and explains skill, knowledge and interest alignment. The official pathway currently ranks six eligible occupations. BLS 2025-35 employment projections are shown separately for occupations with exact SOC mappings; they are labelled as US forecasts, not Malaysian demand. Each result includes a manually verified SDG scenario and target, followed by a four-week plan that prioritizes the user's positive skill gaps and links to checked learning resources, costs, language and time information. Users can track progress and export or restore a plan in English, Chinese or Malay. AI is deliberately narrow: after explicit consent, an OpenAI-compatible Qwen/Qwen3-8B service on RunPod extracts candidate questionnaire fields from redacted CV text. Users review those fields before they affect the profile; the model never calculates scores or rankings. The application uses GenAI API integration, privacy-aware prompt design and cloud deployment skills from the SDC workshops. Replit hosts the public prototype. Local structured catalogues provide occupational, labour, SDG and learning evidence. The core questionnaire-to-plan journey remains usable when the AI service is unavailable.

Word count: 197 / 300

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

Layer	Technology	Role
Interface	Python 3.11+; Streamlit 1.66.0	Questionnaire, results, plans, localization, state
Document	pypdf 6.21.1	Extract selectable text from permitted PDFs; no OCR
AI	RunPod OpenAI-compatible API; Qwen/Qwen3-8B	Optional consent-based CV field extraction
Evidence	O*NET 31.0; BLS 2025-35; UN SDG targets	Career requirements, forecast context, SDG mapping
Storage	Versioned JSON + Streamlit session state	Public catalogues and temporary user state; no account DB
Cloud	Replit	Public Streamlit deployment

1. USER	Questionnaire or synthetic profile; optional text-based PDF CV
	↓
2. STREAMLIT APP	Consent, validation, session state, multilingual UI
	↓
3. DETERMINISTIC ENGINE	O*NET fit, evidence display, SDG cards, four-week plan
	↓
4. OPTIONAL AI	Redacted CV text -> RunPod OpenAI-compatible API -> Qwen/Qwen3-8B
	↓
5. DEPLOYMENT	Replit public app; local JSON evidence catalogues; no account database

Official sponsor resources used: Replit access/credits for deployment and RunPod access/credits for optional model inference. Exact billing tier and region should be confirmed in the platform dashboard before final submission.

3.2 Design-Thinking & Deployment Strategy

The workflow starts from a concrete user job: 'show me plausible paths and the next useful action.' Early testing exposed confusing mixed modes, hidden evidence boundaries, language-state loss and fragile AI retries. The design was iterated into one official pathway with a separate synthetic trend demonstration. Stable career IDs preserve selections across English, Chinese and Malay. Every AI-extracted field requires user review; failures leave the questionnaire usable. Recommendation cards show component scores, coverage and omitted dimensions. Labour evidence is displayed separately when mappings are exact, while aggregate or missing evidence is labelled instead of filled with zero. The learning plan converts only positive gaps into weekly tasks and preserves completed progress across compatible changes. Deployment uses Replit with secrets stored as environment variables. RunPod is replaceable because the client uses an OpenAI-compatible interface. Phase 2 validation will use consented synthetic or non-sensitive profiles, task-completion timing, error logging without CV content, mobile checks and short user interviews. Public-demo instructions will prioritize the questionnaire path so the core experience does not depend on PDF upload or AI availability.

Word count: 171 / 200

4. Preliminary Environmental & SDG Impact Framework

Metric Category	Response
Target SDG Sub-Target(s)	Primary: SDG Target 4.4 - substantially increase the number of youth and adults with relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. Supporting: Target 8.2 (productivity through technological upgrading and innovation) and Target 8.5 (productive employment and decent work).
Current Problem Baseline	Workforce context: 56% of Malaysian professionals reportedly use AI tools, while only 26% have received formal AI training. Product baseline at hackathon start: 0% of pilot users could complete the profile-to-explainable-career-to-four-week-plan journey because no deployed prototype existed.
Core Measurable KPI	Pilot journey completion rate: percentage of consenting test users who create a profile, inspect a ranked career with evidence and SDG contribution, select a career, and generate a four-week plan within five minutes without facilitator intervention.
Phase 2 Impact Target	At least 80% of pilot users complete the defined journey within five minutes. Secondary quality gate: 100% of displayed official careers retain a valid local SDG target/source entry and every generated task identifies its skill gap, time budget and resource status.

Measurement Plan

Use a fixed synthetic profile set plus a small consented usability pilot. Record only step completion, elapsed time, selected language and validation errors; do not retain CV text. Report numerator, denominator and exclusions. The 80% target measures prototype usability and access to actionable guidance, not employment outcomes or national reskilling impact.

Evidence and citations

1. SDC Domain 1 brief, citing Hays Asia Salary Guide 2025 for AI use and formal training figures.
2. SDC Domain 1 brief, citing Malaysia's Ministry of Science, Technology and Innovation for the reskilling estimate.
3. United Nations SDG 4 targets: https://sdgs.un.org/goals/goal4
4. United Nations SDG 8 targets: https://sdgs.un.org/goals/goal8

5. Eligibility & Readiness Checklist

[x]	All team members completed the mandatory GenAI, Database and Cloud Deployment workshops.
[x]	The concept addresses a verified workforce challenge and maps directly to SDG 4.4, with supporting links to SDG 8.2 and 8.5.
[x]	The submission is a functional deployed application, not a theoretical business case.
[x]	All fields are completed within the stated word limits.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Yes / No	No
Details	The submitted implementation was created during the SDC 2026 hackathon period. Third-party foundations are declared below and in the Technical Tools Declaration.

6.2 Third-Party Code, Datasets, or Models

Yes / No	Yes

Name	Type	License / Terms
Streamlit 1.66.0	Code framework	Apache-2.0
pypdf 6.21.1	Code library	BSD-3-Clause
O*NET 31.0	Public dataset	CC BY 4.0; USDOL/ETA attribution
BLS Employment Projections 2025-35	Public government data	US government publication; source attributed
Qwen/Qwen3-8B	Pre-trained model	Provider/model terms apply; served through RunPod
UN SDG target text	Public reference data	Source attributed; project mappings are not UN endorsements
Learning-resource metadata	Public web metadata	Links and provider names attributed; course content remains with providers

6.3 Employment-Related IP Conflicts

Yes / No	No
Details	The sole team member is submitting as a student and has identified no employment, internship or consulting agreement that claims ownership of this work.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project is original work created during the SDC 2026 hackathon period.

Team Leader	YAO Yunshu
Signature	_____
Date	24 October 2026

Final action before upload: sign the line above and confirm the sponsor billing tier/region in Section 3.1. All other fields are populated.